p-adic Analysis 3

Siddharth K

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Problem 1

Part (a)

The sequence n^n does not converge because, for any p and any $\varepsilon = p^{-k}$, we can find $n > k$ relatively prime to p, so that $v_p(n^n) = 0$.

Part (b)

The p-adic valuation of $1/n!$ tends to $-\infty$, so the p-adic distance goes to ∞ and the sequence diverges.

Part (c)

We claim that $(1+p)^{p^n} \to 1$. Expanding gives us

$$(1+p)^{p^n} - 1 = \sum_{k=1}^{p^n} \binom{p^n}{k} p^k.$$

We have that

$$\binom{p^n}{k} = \frac{p^n}{k}\binom{p^n-1}{k-1},$$

$$v_p\left(\binom{p^n}{k}\right) = n - v_p(k) + v_p\left(\binom{p^n-1}{k-1}\right) \geq n - v_p(k),$$

$$v_p\left(\binom{p^n}{k}p^k\right) \geq n - v_p(k) + k.$$

For $k = 1$, the term $\binom{p^n}{1}p$ has valuation $n+1$. For $k \geq 2$, since $v_p(k) \leq k-1$,

$$n - v_p(k) + k \geq n + 1.$$

So

$$v_p((1+p)^{p^n} - 1) = n + 1 \to \infty,$$

and the sum converges.

Part (d)

If p is odd,

$$p-1=-(1-p),$$
$$(p-1)^{p^n}=-(1-p)^{p^n}\to 1,$$

by the same reasoning as part (c). Therefore,

$$(p-1)^{p^n}\to -1.$$

If p is even (2),

$$(p-1)^{p^n}=1\to 1.$$

Problem 4

Part (a)

We find two subsequences converging to different limits.

First, consider the sequence $\frac{1}{p^n-1}$. We claim that it converges to -1. We have that

$$v_p\left(\frac{1}{p^n-1}+1\right)=v_p\left(\frac{p^n}{p^n-1}\right)=n\to\infty,$$

so the sequence converges to -1.

Now, consider any integer sequence a_n satisfying

$$2a_n\equiv -1 \pmod{p^n}.$$

Assuming p is odd, such a sequence always exists. Furthermore, $v_p(a_n)=0 \pmod{p^n}$. So

$$2a_n+1\equiv\frac{2a_n+1}{a_n}=2+\frac{1}{a_n}\equiv 0 \pmod{p^n},$$

which means that $\frac{1}{a_n}\to -2$.

Part (b)

Fix $x\in\mathbb{Q}_p$. We can write $x=\frac{y}{p^k}$ for some integer k. Fix $N>0$. We wish to find some n such that

$$v_p\left(\frac{y}{p^k}-\frac{1}{n}\right)>N.$$

Choose n such that

$$y\cdot n\equiv p^k \pmod{p^N},$$

which always exists because $\gcd(y,p^k)=0$. Then we have that

$$\frac{y\cdot n-p^k}{p^k\cdot n}\equiv 0 \pmod{p^N},$$

so $v_p(x,1/n)>N$.

Problem 7

Let $a_n \to a$ in $\mathbb{Q}_p$. If $a = 0$, $|a_n|_p = |a_n - 0|_p \to 0$. If $a \neq 0$, then $|a|_p > 0$. Since $|a_n - a|_p \to 0$, there exists N such that $n \geq N$ implies

$$|a_n - a|_p < |a|_p.$$

Now write $a_n = a + (a_n - a)$. For $n \geq N$,

$$|a|_p > |a_n - a|_p,$$

since $|a_n - a|_p \to 0$. By the ultrametric triangle inequality,

$$|a_n|_p = |a + (a_n - a)|_p \leq \max(|a|_p, |a_n - a|_p) = |a|_p.$$

Problem 10

Choose a prime $q_n > n!^2$ and let m_n be the nearest integer to $\frac{q_n \pi}{n!}$. Then define

$$a_n = \frac{n! \cdot m_n}{q_n}.$$

We claim that $a_n \to \pi$ under the usual metric. Since m_n is the nearest integer to $q_n\pi/n!$,

$$\left| m_n - \frac{q_n \pi}{n!} \right| \leq \frac{1}{2}.$$

We have that

$$|a_n - \pi| = \left| \frac{n! \cdot m_n}{q_n} - \pi \right| = \frac{n!}{q_n} \left| m_n - \frac{q_n \cdot \pi}{n!} \right| \leq \frac{n!}{2q_n} < \frac{1}{2n!} \to 0,$$

since $q_n > n!^2$.

Now, fix a prime p. For sufficiently large n, all $q_n > p$ (since q_n is prime), so $v_p(q_n) = 0$. So we have that

$$v_p(a_n) = v_p(n!) + v_p(m_n) - v_p(q_n) \geq v_p(n!) \to \infty.$$

Therefore, $|a_n|_p \to 0$.